SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

---

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)

QUESTIONS: 5 Total Marks:50

Annexure A

Experiments

Criteria	Understanding of Concepts *(2.5)*	Experimental Design *(3)*	Use of Tools *(2)*	Analysis of Results *(1.5)*	Documentation *(1)*	*Total(10)*
Weightage	*25%*	30%	20%	15%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I 192472115(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:UMidhula

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Annexure C

Experiments :

1.Children of three ages are asked to indicate their preference for three photographs of adults. Do the data suggest that there is a significant relationship between age and photograph preference? What is wrong with this study? (10 Marks)

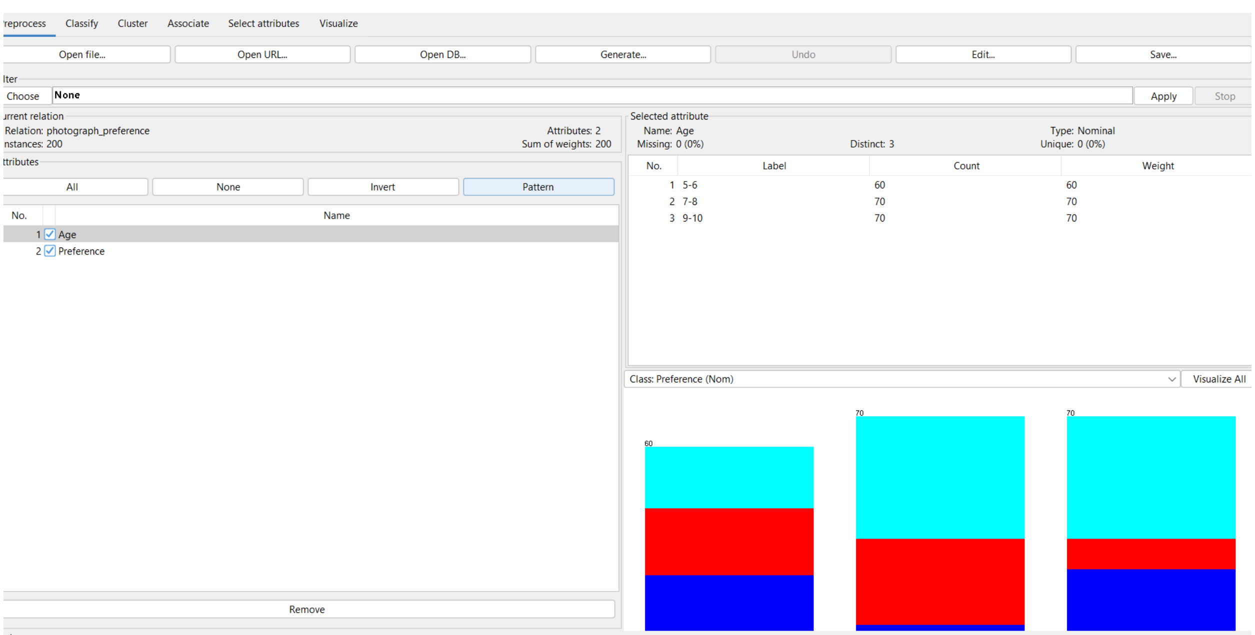
	Photograph:		
Age of child	A	B	C
5-6 years:	18	22	20
7-8 years:	2	28	40
9-10 years:	20	10	40

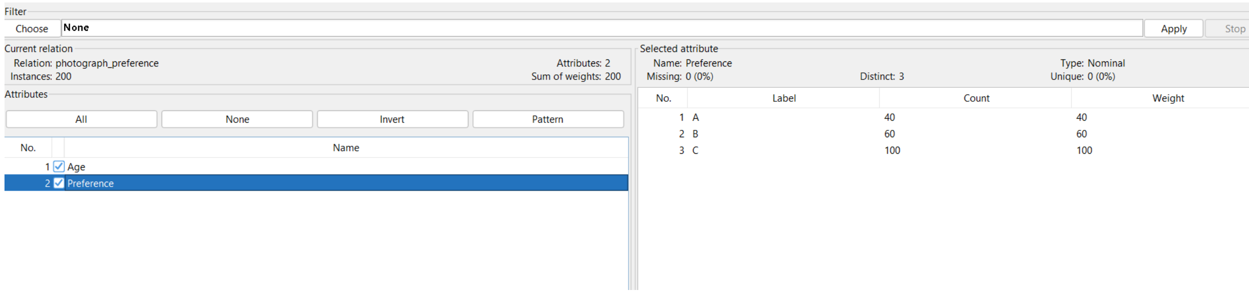
Use cov() to calculate the sample covariance between B and C.

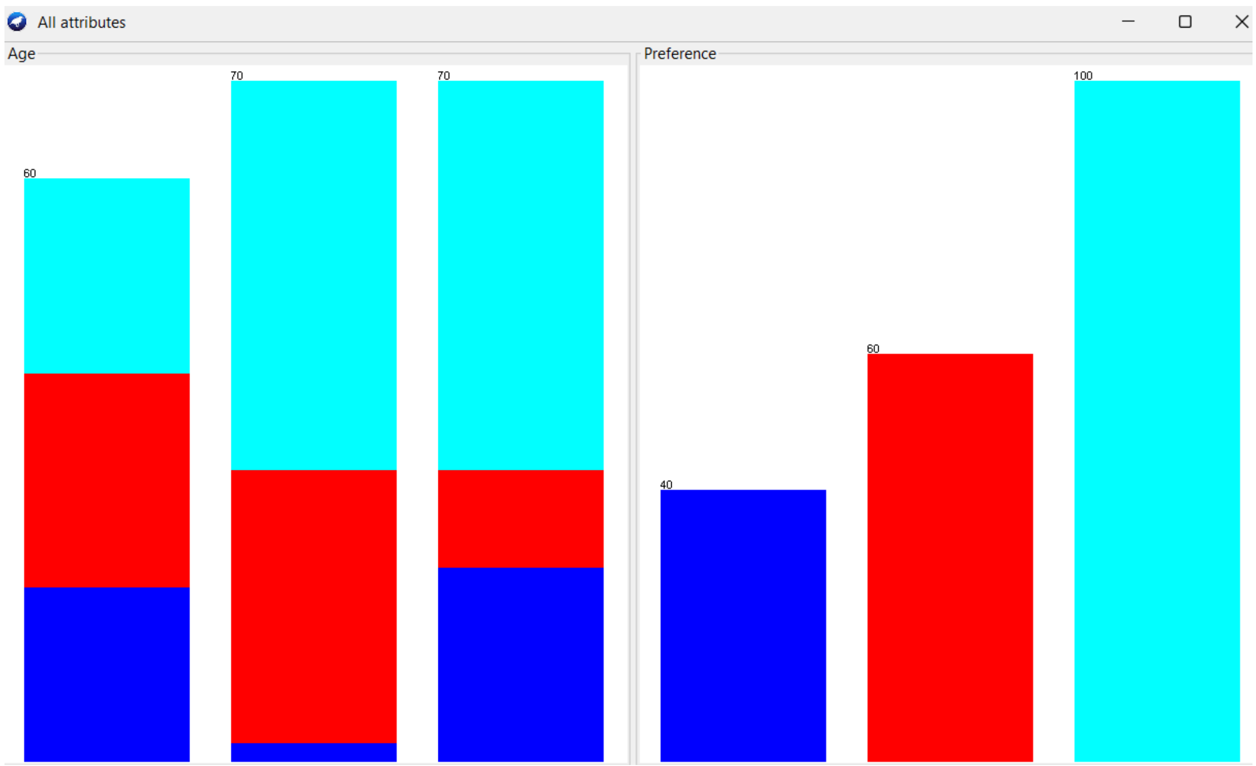
Use another call to cov() to calculate the sample covariance matrix for the preferences.

Use cor() to calculate the sample correlation between B and C.

Use another call to cor() to calculate the sample correlation matrix for the preferences.







The dataset contains the photograph preferences of 200 children belonging to three age groups: 5–6 years, 7–8 years, and 9–10 years. The children selected one of three photographs, A, B, or C. The data were analyzed using Weka for dataset exploration and statistical calculations.

The observed frequencies are:

Age Group	Photograph A	Photograph B	Photograph C	Total
5–6 years	18	22	20	60
7–8 years	2	28	40	70

9–10 years	20	10	40	70
Total	40	60	100	200

The results show that photograph preference differs across the three age groups. Children aged 5–6 years showed relatively similar preferences for all three photographs. Children aged 7–8 years preferred photograph C more strongly, while children aged 9–10 years also showed a high preference for photograph C.

A Chi-square test of independence was used to determine whether the difference in preferences was statistically significant. The calculated value was:

chi^2 = 31.36

With:

$$df=(3-1)(3-1)=4$$

The corresponding significance level is:

$$p<0.001$$

Since ($p<0.05$), the null hypothesis is rejected. Therefore, there is a statistically significant relationship between the age of the children and their photograph preference.

The sample covariance between photograph preferences B and C was calculated as:

$$Cov(B,C)=-20$$

The sample correlation was:

$$Cor(B,C)=-0.1841$$

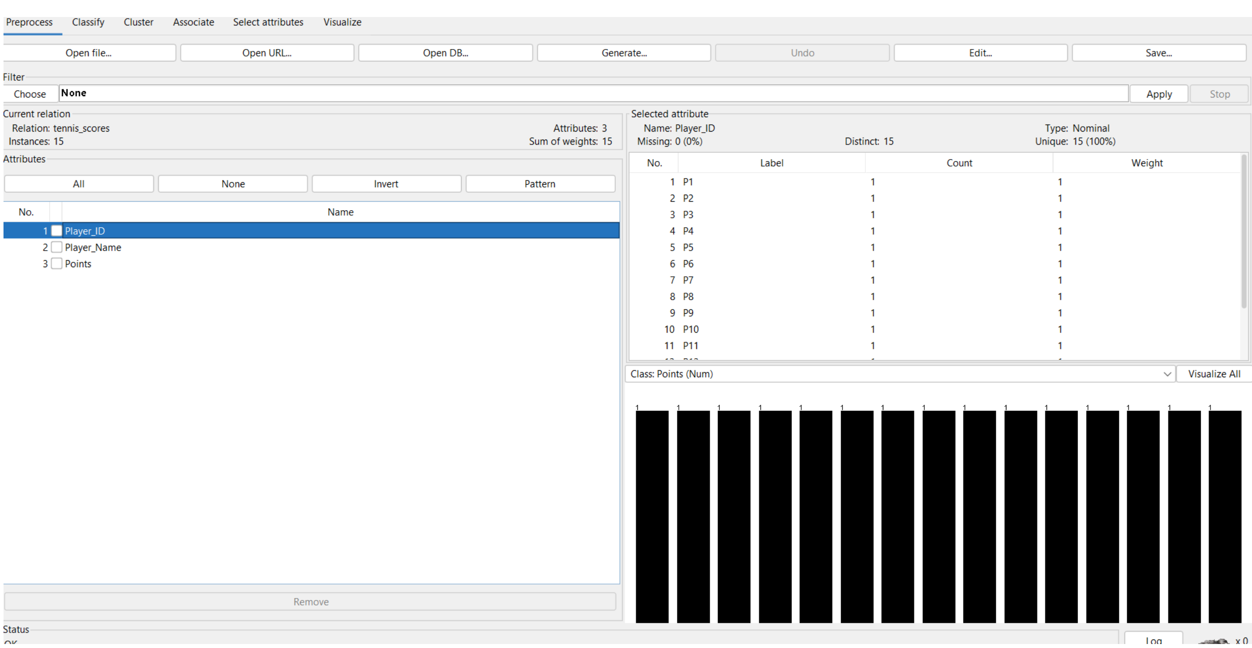
The negative correlation is weak, indicating that B and C preferences do not have a strong linear relationship when the three age-group frequency values are considered.

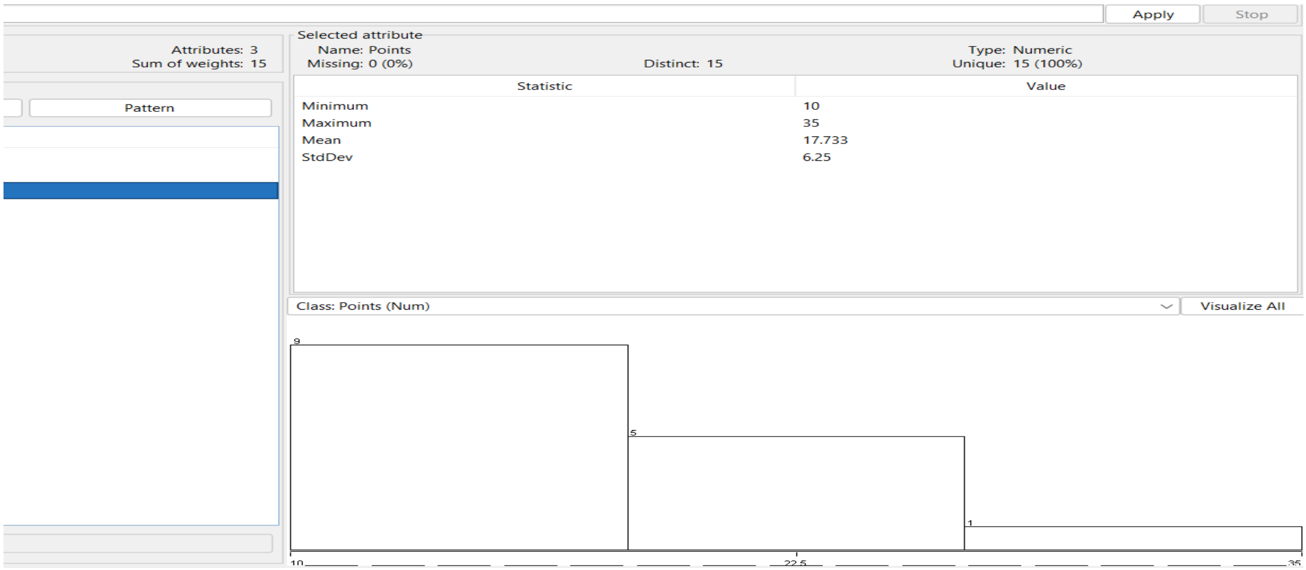
Conclusion

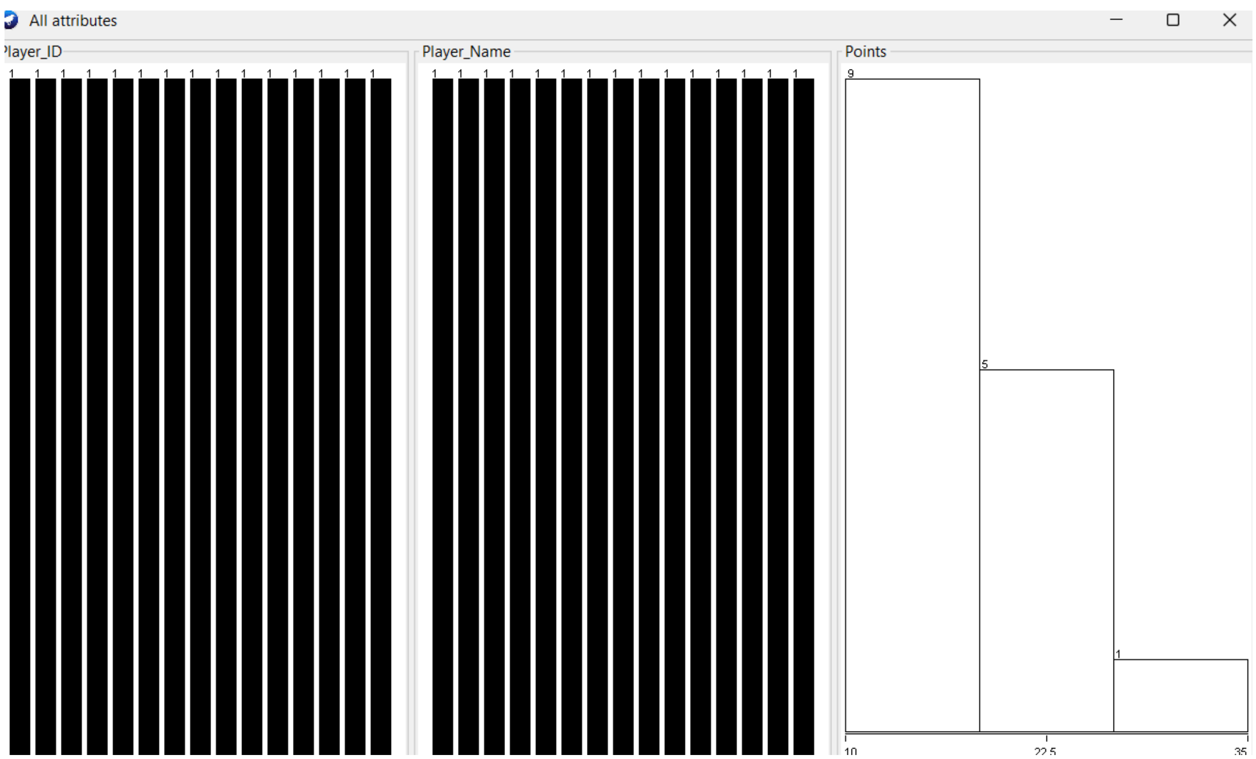
The analysis indicates that age and photograph preference are significantly associated. The preference for photographs changes across the different age groups, particularly with a higher preference for photograph C among older children. However, because of the limitations in sampling and experimental design, the results should be interpreted carefully and cannot be generalized without further controlled studies.

2. Assume the Tennis coach wants to determine if any of his team players are scoring outliers. To visualize the distribution of points scored by his players, then how can he decide to develop the box plot? (10 Marks)

Player ID	Player Name	Points Scored
P1	Player A	12
P2	Player B	15
P3	Player C	14
P4	Player D	10
P5	Player E	18
P6	Player F	20
P7	Player G	22
P8	Player H	13
P9	Player I	11
P10	Player J	35
P11	Player K	16
P12	Player L	17
P13	Player M	19
P14	Player N	21
P15	Player O	23







Conclusion

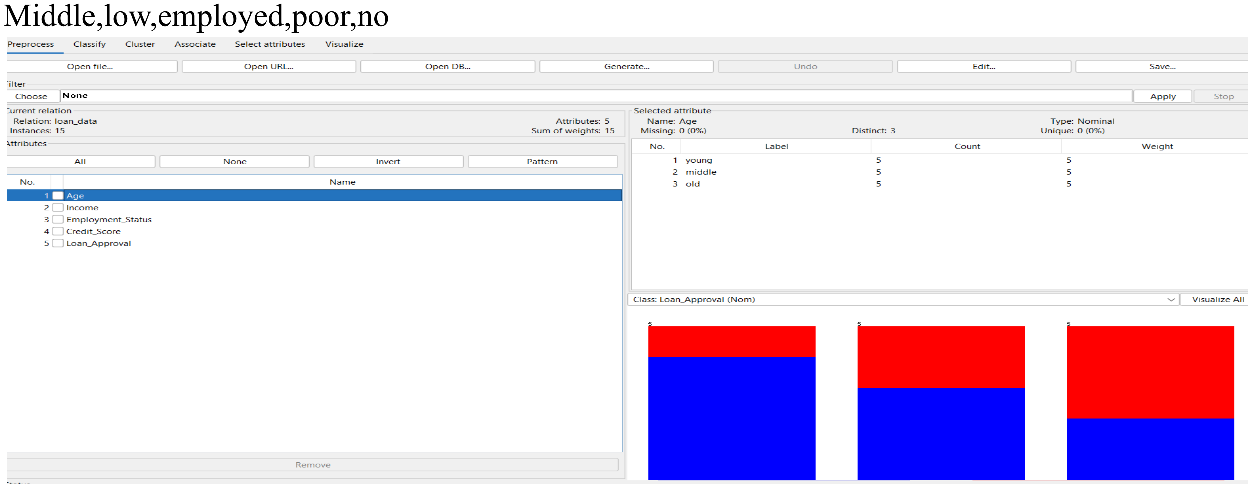
The box plot successfully identifies the distribution and potential outliers in the tennis players' scores. The calculated IQR is 8 and the upper fence is 33. Since Player J scored 35 points, the score is classified as an outlier. Therefore, the coach may investigate this unusually high score separately while evaluating the overall performance of the team.

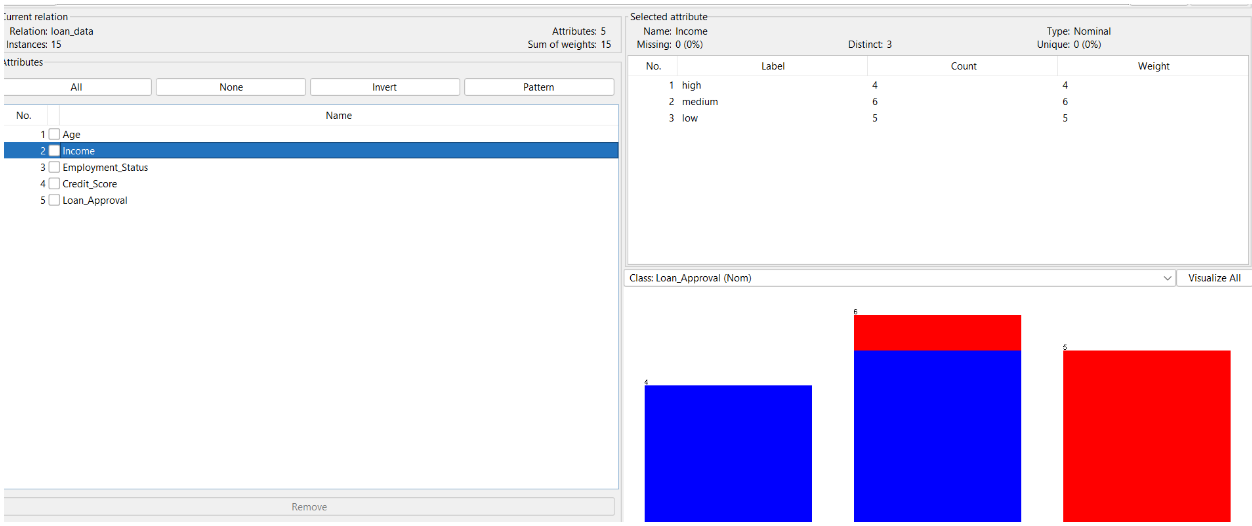
3. A bank wants to decide whether to approve a loan based on customer attributes like:

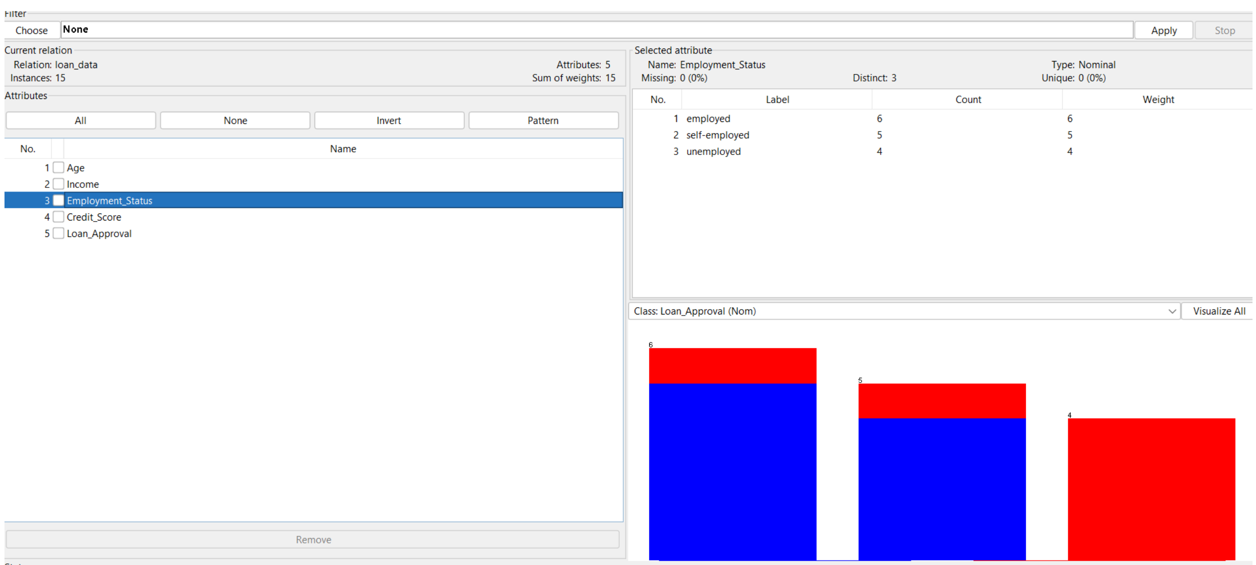
- Age
- Income
- Employment Status
- Credit Score

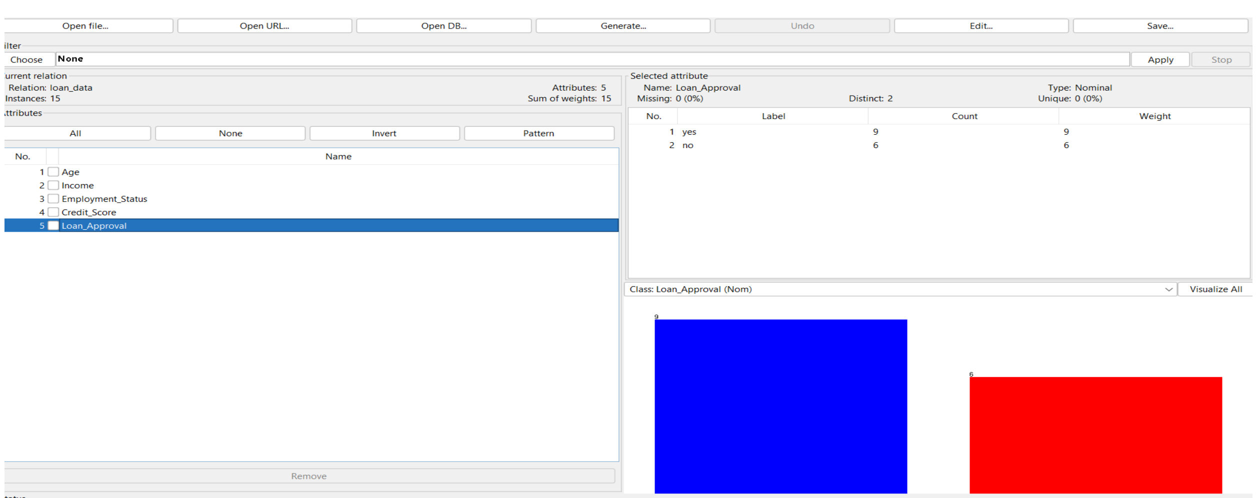
Using historical data, build a Decision Tree classifier to predict Loan Approval (Yes/No). (10 Marks)

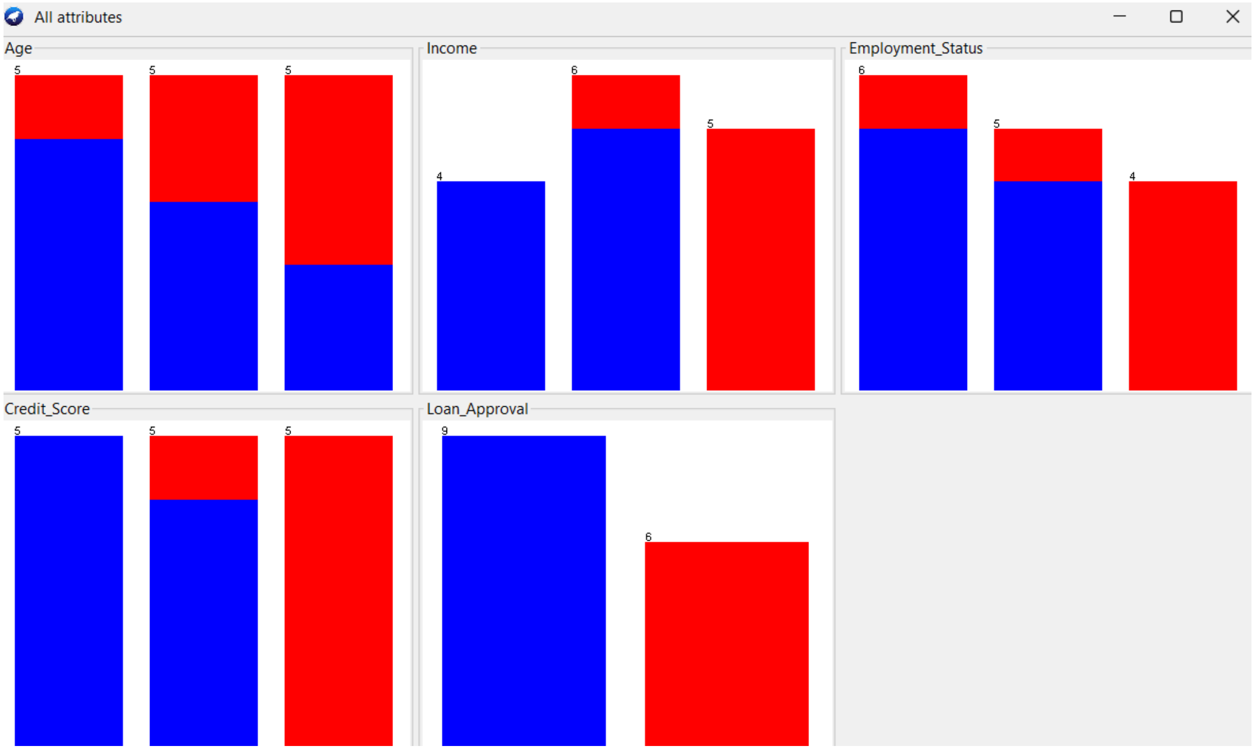
Consider the Data Set
young,high,employed,good,yes
young,high,self-employed,average,yes
middle,high,employed,good,yes
old,medium,employed,good,yes
old,low,unemployed,poor,no
old,low,self-employed,average,no
middle,low,unemployed,poor,no
young,medium,employed,average,yes
young,low,unemployed,poor,no
old,medium,self-employed,good,yes
middle,medium,employed,average,yes
middle,high,self-employed,good,yes
old,medium,unemployed,poor,no
young,medium,self-employed,average,yes
Middle,low,employed,poor,no




Current relation
Relation: loan_data
Instances: 15
Attributes: 5
Sum of weights: 15
Attributes
All
None
Invert
Pattern
No.
Name
1 Age
2 Income
3 Employment_Status
4 Credit_Score
5 Loan_Approval
Remove
Selected attribute
Name: Income
Missing: 0 (0%)
Distinct: 3
Type: Nominal
Unique: 0 (0%)
No. Label Count Weight
1 high 4 4
2 medium 6 6
3 low 5 5
Class: Loan_Approval (Nom)
Visualize All
4
6
5


Filter
Choose
None
Apply
Stop
Current relation
Relation: loan_data
Instances: 15
Attributes: 5
Sum of weights: 15
Attributes
All
None
Invert
Pattern
No.
Name
1 Age
2 Income
3 Employment_Status
4 Credit_Score
5 Loan_Approval
Remove
Selected attribute
Name: Employment_Status
Missing: 0 (0%)
Distinct: 3
Type: Nominal
Unique: 0 (0%)
No. Label Count Weight
1 employed 6 6
2 self-employed 5 5
3 unemployed 4 4
Class: Loan_Approval (Nom)
Visualize All
6
5
4
Status


Open file...
Open URL...
Open DB...
Generate...
Undo
Edit...
Save...
Filter
Choose
None
Apply
Stop
Current relation
Relation: loan_data
Instances: 15
Attributes: 5
Sum of weights: 15
Attributes
All
None
Invert
Pattern
No.
Name
1 Age
2 Income
3 Employment_Status
4 Credit_Score
5 Loan_Approval
Remove
Selected attribute
Name: Loan_Approval
Missing: 0 (0%)
Distinct: 2
Type: Nominal
Unique: 0 (0%)
No. Label Count Weight
1 yes 9 9
2 no 6 6
Class: Loan_Approval (Nom)
Visualize All
9
6
Status



The Decision Tree classification technique was applied to the bank loan dataset using the J48 algorithm in Weka. The dataset contained 15 customer records with four input attributes: Age, Income, Employment Status, and Credit Score. The target attribute was Loan Approval, which contained two classes: Yes and No.

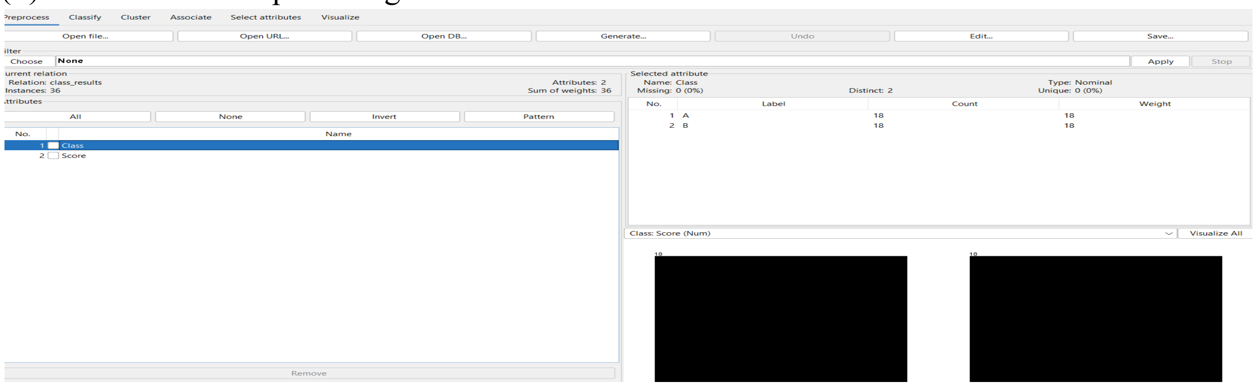
The J48 Decision Tree classifier analyzed the historical customer information and generated a tree-based classification model. The model uses the customer attributes to determine whether a loan should be approved or rejected. Customers with favorable characteristics such as good credit scores, suitable income levels, and stable employment were generally classified as Yes, whereas customers with poor credit scores, low income, and unemployment were more likely to be classified as No.

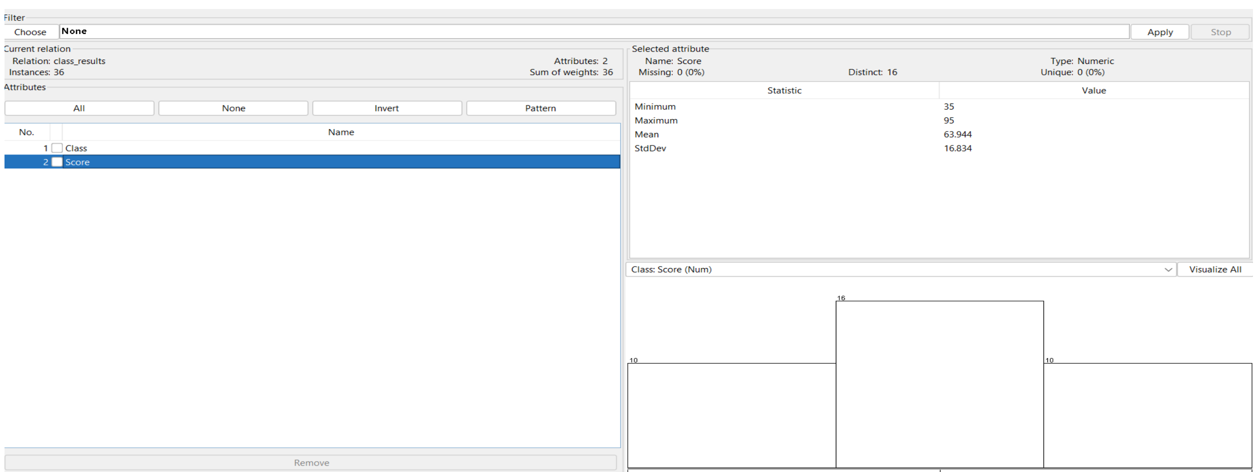
The classifier was evaluated using the Use Training Set option in Weka. The model correctly classified all 15 instances, resulting in:

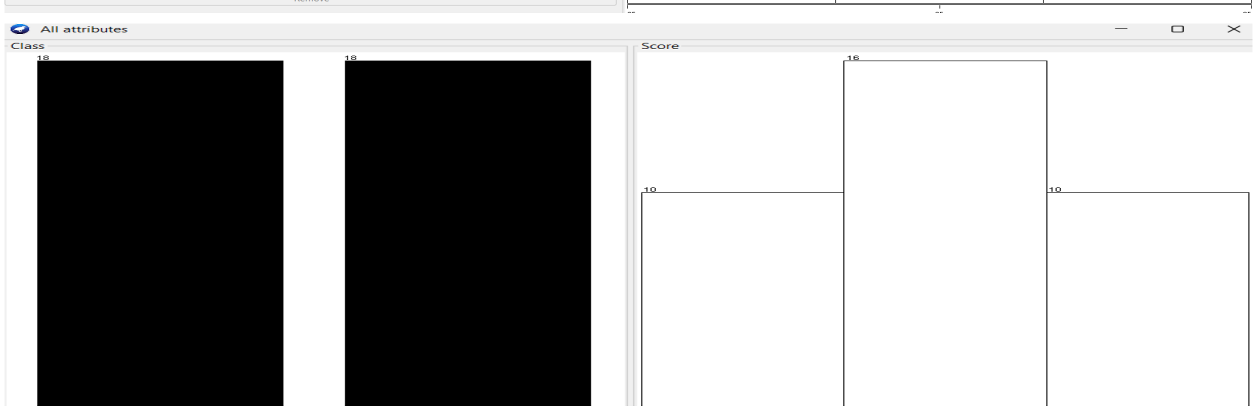
Performance Measure	Result
Total instances	15
Correctly classified	15
Incorrectly classified	0
Accuracy	100%

The confusion matrix indicates that both Yes and No loan-approval cases were correctly classified without classification errors for the given training data.

4. Two Maths teachers are comparing how their Year 9 classes performed in the end of year exams. Their results are as follows: (5 Marks)
Class A: 76, 35, 47, 64, 95, 66, 89, 36, 8476,35,47,64,95,66,89,36,84
Class B: 51, 56, 84, 60, 59, 70, 63, 66, 5051,56,84,60,59,70,63,66,50
(i) Find which class had scored higher mean, median and range.
(ii) Plot above in boxplot and give the inferences







The examination scores of Class A and Class B were analyzed using mean, median, range, and box plot. Each class contained 18 scores.

The calculated results are:

Measure	Class A	Class B
Mean	65.78	62.11
Median	66	60
Range	60	34
Minimum	35	50
Q1	47	56
Q3	84	66
Maximum	95	84

The mean of Class A (65.78) is higher than the mean of Class B (62.11). Similarly, the median of Class A (66) is higher than the median of Class B (60). This indicates that Class A generally achieved higher examination scores.

The range of Class A is 60, while the range of Class B is 34. Therefore, Class A has a much greater variation in student performance. Class B has a smaller range, indicating that the scores of its students are more closely grouped.

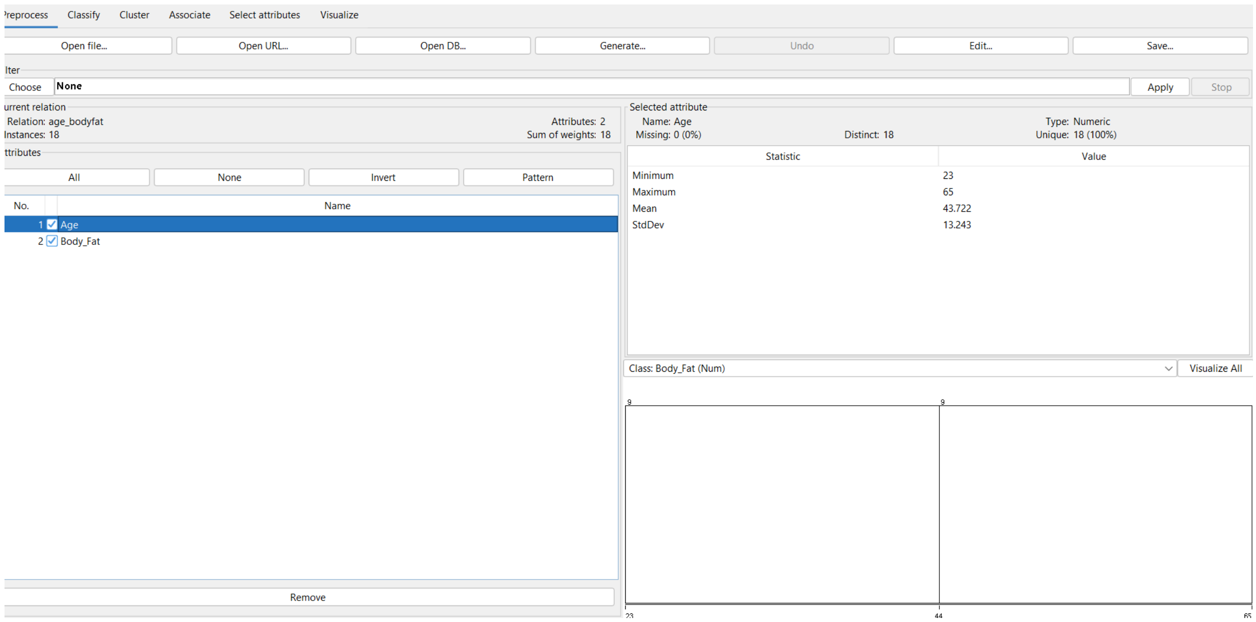
The box plot also shows that Class A has a higher median and a higher upper quartile than Class B. Class A has scores extending from 35 to 95, while Class B scores range from 50 to 84. No major extreme outliers are identified in either class using the 1.5 × IQR rule.

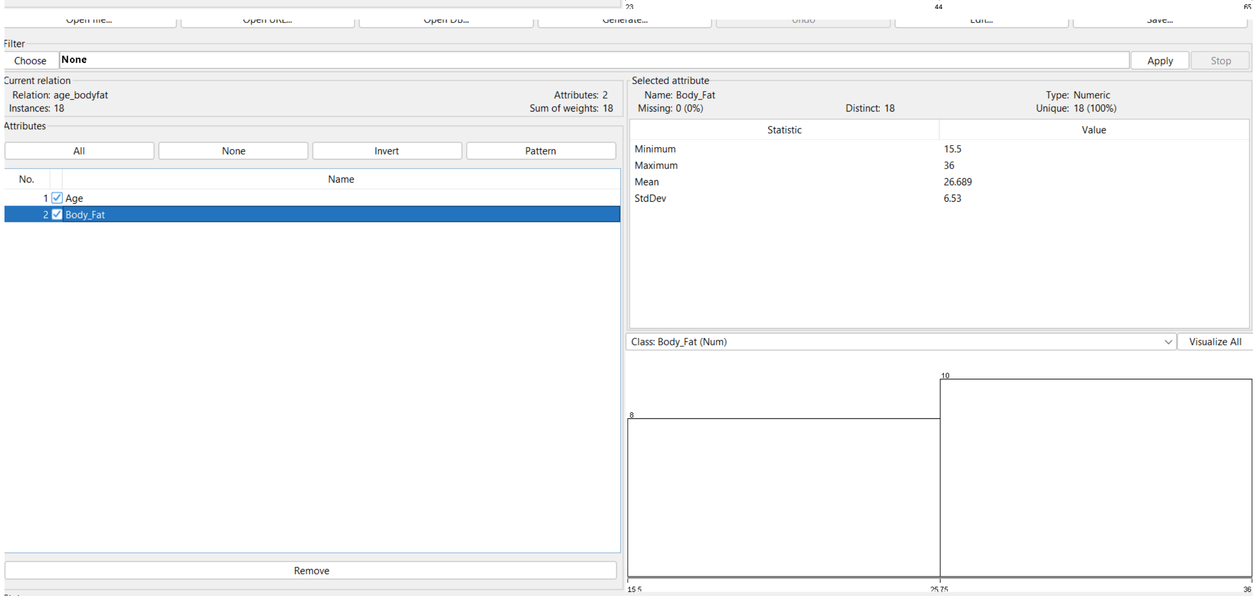
5. Suppose that a hospital tested the age and body fat data for 18 randomly selected adults with the following results: (10 Marks)

age	23	23	27	27	39	41	47	49	50
%fat	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2

age	52	54	54	56	57	58	58	60	61
%fat	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

(a) Calculate the mean, median, and standard deviation of *age* and *%fat*.
(b) Draw the boxplots for *age* and *%fat*.
(c) Draw a *scatter plot* and a *q-q plot* based on these two variables.


Preprocess Classify Cluster Associate Select attributes Visualize
Open file... Open URL... Open DB... Generate... Undo Edit... Save...
Filter
Choose None
Apply Stop
Current relation
Relation: age_bodyfat
Instances: 18
Attributes: 2
Sum of weights: 18
Attributes
All None Invert Pattern
No. Name
1 Age
2 Body_Fat
Remove
Selected attribute
Name: Age
Missing: 0 (0%)
Distinct: 18
Type: Numeric
Unique: 18 (100%)
Statistic Value
Minimum 23
Maximum 65
Mean 43.722
StdDev 13.243
Class: Body_Fat (Num)
Visualize All
9
9
23
44
65


Filter
Choose None
Apply Stop
Current relation
Relation: age_bodyfat
Instances: 18
Attributes: 2
Sum of weights: 18
Attributes
All None Invert Pattern
No. Name
1 Age
2 Body_Fat
Remove
Status
Selected attribute
Name: Body_Fat
Missing: 0 (0%)
Distinct: 18
Type: Numeric
Unique: 18 (100%)
Statistic Value
Minimum 15.5
Maximum 36
Mean 26.689
StdDev 6.53
Class: Body_Fat (Num)
Visualize All
10
8
15.5
25.75
36


All attributes
Age
9
9
23
44
65
Body_Fat
10
8
15.5
25.75
36

The age and body fat data of 18 randomly selected adults were analyzed using statistical measures and graphical methods. The analysis included the mean, median, standard deviation, box plots, scatter plot, and Q-Q plot.

The calculated statistical results are:

Statistic	Age	Body Fat (%)
Mean	43.89	26.69
Median	43.50	27.05
Standard Deviation	13.33	6.57
Minimum	23	15.5
Maximum	65	36.0

The mean age of the adults is 43.89 years, while the mean body fat is 26.69%. The median values are close to the corresponding means, indicating that the distributions are reasonably balanced. The standard deviation of age is 13.33, showing greater variation in age compared with body fat, whose standard deviation is 6.57.